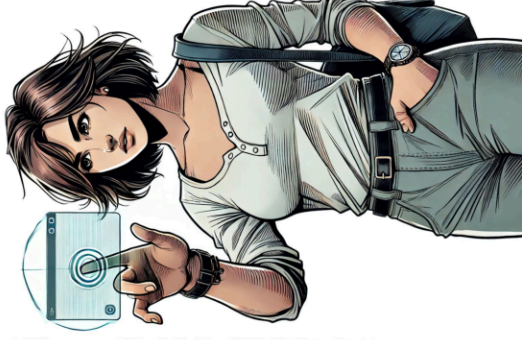


Class 14

Ethics and Accessibility: Real-World Cases

Ethical Design
Invisible Design
Inclusive Design
Consent and Privacy
Dark Patterns



1

In this class we will...

- We'll explore how **ethical issues** show up in real UX decisions.
- You'll analyze **real cases** where design caused harm, and how it could be fixed.
- We'll focus on **consent, accessibility, bias, and inclusion**, all from an engineering leadership lens.



2



LinkedIn® – The Disappearing Checkbox

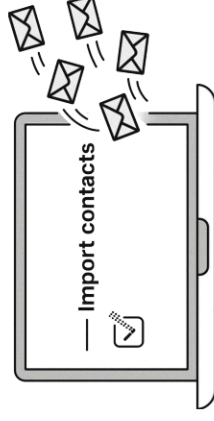
- Story:
 - Back on 2011, users signed up for LinkedIn, using a form that invited to import their email contacts. Hidden in the flow was a **pre-checked box** that allowed the platform to automatically **email everyone** in the user's address book.
- Consequences:
 - Many users didn't realize they were giving consent. As a result, **millions of unwanted invitations were sent in their name**, damaging personal and professional relationships.
 - The **class-action lawsuit** was filed in 2013, and the case was settled in 2015, when LinkedIn agreed to pay **\$13 million** to avoid further litigation.

3



LinkedIn® – Violating Consent and Autonomy

- It **violates user autonomy**: decisions are made for users without clarity.
- Prioritizes **marketing KPIs** over user rights.
- Sets a **precedent** for manipulation in microinteractions.
- It's a **dark pattern**: design **used to trick, not serve**.



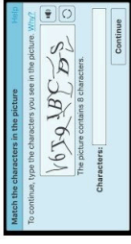
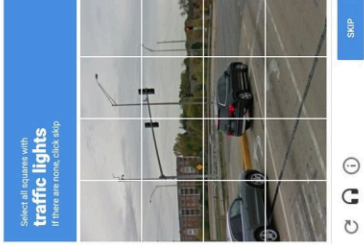
4



Google – CAPTCHA Exclusion

Jorge Maturana, 2025

- Story:
 - To protect websites from bots, Google's reCAPTCHA asked users to solve **visual puzzles**: click on all the traffic lights, or decipher blurry text. Users with visual, motor, or cognitive disabilities, these tasks were nearly impossible. Even the **audio version** failed to help many.
- Consequences:
 - Millions of users **were blocked** from accessing services like healthcare, education, or government platforms.



5



Google – Designing Inclusive CAPTCHA Alternatives

Jorge Maturana, 2025

- 2014: No CAPTCHA reCAPTCHA:
 - “I’m not a robot” checkbox
 - Replaced distorted text with a simple click, using behavior analysis in the background.
- 2018: reCAPTCHA v3 (no puzzles at all)
 - Assigns a “**trust**” score to the user based on their behavior on the site. The system decides whether or not to display additional verification.
 - **Invisible to the user.** Does not block access. Reduces friction and exclusion.
- Trends:
 - Honeypots (hidden fields that bots fill out, but humans don’t)
 - Typing time and rhythm analysis
 - Progressive authentication: basic access first, then identity verification based on behavior.



6

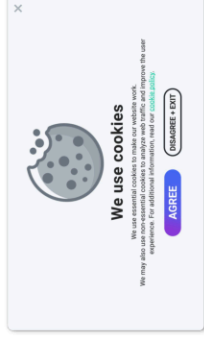
• Story:

- To comply with GDPR (General Data Protection Regulation, a European law), many websites used IAB Europe's TCR (**Transparency and Consent Framework**), but instead of offering real choice, they created **Consent Walls**: users had to accept all cookies to access content.
- It pressures users to accept tracking they don't understand: Rejecting cookies was confusing or hidden.

• Consequences:

- What looked like compliance was actually manipulation, and trust was lost.
- In 2022, **Belgium's Data Protection Authority ruled this approach illegal. IAB Europe was fined** and forced to revise its framework.

7



amazon – AI Bias in Hiring

• Story:

- To speed up hiring, Amazon built an **AI to screen résumés**. It learned from past hiring data — but that data was **biased towards men**.
- Soon, the AI began **rejecting CVs** with words like “women’s chess club” or graduates from all-female colleges. It **favorred male-coded language and patterns**.

• Consequences:

- In 2018, Amazon shut it down. What seemed like smart automation became **silent discrimination** — built not by malice, but by unexamined data and invisible design.
- Led to use diverse and balanced datasets and include humans in the decision loop to make results explainable.

8



amazon – Invisible design

Jorge Maturana, 2025

- How this case relates to UX/UI?
 - UX is **not just screens**, it's the **full experience**.
 - When an AI rejects a user silently, that's a **designed interaction**, even if it's not visual.
 - Users had no feedback, no transparency, no control.
 - **Ethical UX** means asking: Is this experience fair, clear, and inclusive? (even behind the scenes)
- **Invisible design**:
 - **Decisions that shape the user experience without being directly visible on the screen.**
"It includes rules and systems that quietly control how a system reacts to the user, often silently."
 - **Examples**: Recommendation algorithms (Netflix, YouTube, TikTok), Dynamic pricing (Uber, Amazon, airlines), Search engine result order, Notifications and alerts, Content Modulation on Social Media, Hidden Limits in Banking Systems.

9



RYANAIR – Hidden Costs at Checkout

Jorge Maturana, 2025

- **Story**:
 - Ryanair displayed **ultra-low ticket prices to attract travelers**. But as users progressed through checkout, hidden costs began to surface: baggage fees, seat selection, payment charges.
 - Many **add-ons were preselected or hard to decline**. The interface nudged users toward spending more without realizing it.
- **Consequences**:
 - Users felt **tricked and frustrated**.
 - 2018-2020: Consumer watchdogs in Europe investigated and **sanctioned** the airline.
 - Ryanair became a **textbook example of dark patterns** in e-commerce.

10



Bloomberg Terminal: Overload for Neurodiverse Users

- Financial dashboards like Bloomberg Terminal (25.000 USD/year, designed in 1982) **overload** with data and motion.
- **Neurodivergent users** (ADHD, autism) may feel overwhelmed or anxious.
- Even typical users can suffer from **attention fatigue**.
- **Complexity wasn't reduced**, but new efforts include more **guided onboarding** and cleaner interfaces in the web version of Bloomberg.



11



How complex a UI should be?

- William Ross Ashby:
 - **Law of Requisite Variety:** To effectively manage a complex system, the interface must offer enough complexity to reflect and control that system, but not more.
- **Balancing Act:** Designing for Control and Clarity:
 - **Expose only necessary complexity**
 - Reveal advanced functions **progressively**. Offer **shortcuts** for expert users and **guidance** for novices (Hick's Law).
 - **Support cognitive diversity**
 - Allow **personalization** of layout, themes, font sizes. Use calm visual design, reduce **simultaneous stimuli**.
 - **Chunk information**
 - **Break data** into digestible sections, use tabs, collapsible panels, or step-by-step flows (Hick's Law, Gestalt).
 - **Use visual hierarchy and structure**
 - Prioritize what's actionable, **Group** related controls and use consistent patterns (Gestalt)
 - **Train and onboard effectively**
 - Provide context-sensitive **help** and **walkthroughs**, Use **examples** and common use case **templates**.



12

Startup Teams – Ethical Engineering Dilemmas

- Story:
 - In the fast-paced world of **startups**, teams often rush to release features and impress investors. UX and engineering decisions are made quickly, and sometimes **ethically compromised**.
 - **Dark patterns** are used to boost engagement and data is collected without consent.
 - Team members raise concerns, but are told: “*Let’s fix it later, we need growth now*”.
- Consequences:
 - Loss of **user trust** and reputational damage.
 - **Legal risk** (e.g., data privacy violations).
 - **Internal tension** and designer burnout.
 - **Ethical erosion** becomes product and UX debt.



Guidelines for Ethical Decisions

- Is the user **informed**?
 - Clear, honest language.
 - No hidden terms or tricks.
- Does the user have **control**?
 - Easy to say yes or no.
 - Can edit or delete data.
- Is it fair and **respectful**?
 - Helps the user, not just the business.
 - No manipulation or pressure.
- Is it **inclusive**?
 - Works for everyone, including people with disabilities or different ways of thinking.
- Is the **data use ethical**?
 - Collect only what’s needed.
 - Be clear about how it’s used.

Would you feel comfortable presenting the system in public, to your friends or family?





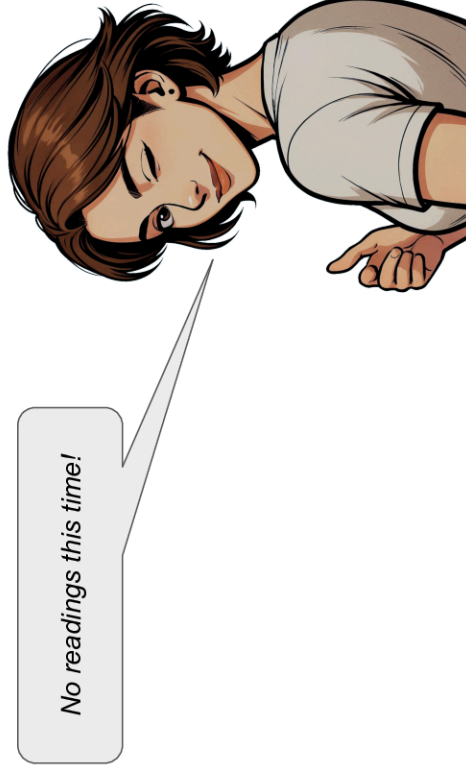
Summary

- UX choices have real-world impact, ethically and legally.
- Dark patterns and biased designs erode trust.
- Accessibility is essential, not optional.
- Invisible design can cause hidden harm.
- Ethical UX means clarity, fairness, and inclusion.

15



Readings



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